Augmented Situation Awareness: Reducing Cognitive Load in Campus Security via Spatially-Anchored AR Visualization

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Abstract—Large-scale campus monitoring systems can quickly generate thousands of alerts in a short time. As these systems detect anomalies, human operators must make sense of these alerts and respond physically. Conventional dashboards often present streams of text or other visual summaries such as markers on a geographic display that require operators to interpret these abstractions into physical locations.

This work evaluates the potential of spatially anchored Augmented Reality (AR) visualization to offload cognitive translation of spatial information from the operator. We introduce a light-weight AR client that interprets standardized event streams and renders detected anomalies as location-bound visual overlays on the operator's view. The contribution of this work lies in the description of the interface layer between automated event detection and operator response.

In a controlled within-subjects user study ($N = 20$), participants using an AR interface needed 42% less time to accomplish the multi-point campus navigation task than with traditional dashboard displays. Additionally, NASA-TLX results suggest that the AR interface was associated with significantly reduced mental workload and frustration compared to 2D interfaces.

These results suggest that spatial computing interfaces can substantially reduce interpretation latency between automated anomaly detection and human operational response.

Index Terms—Spatial Cognitive Offloading, AR Visualization, Situation Awareness, Perceptual Abstraction, Interpretation Latency, Ergonomic Interface Design.

I. Introduction

The main point of this paper is that the visualizations that incorporate spatial positioning reduce the cognitive distance between the symbols on the screen and the actions that need to be taken. The monitoring systems in the majority of the cases are equipped with algorithms that allow them to detect anomalies automatically. However, despite an abundance of automation, the act of alerting and taking action on the detected discrepancies remains a responsibility of humans. Therefore, the issue of visualization becomes one of the main concerns regarding such systems.

A. The Cognitive Cost of Symbolic Monitoring Interfaces

Conventional dashboard interfaces present alerts as textual streams or static map markers. Operators must continuously perform mental spatial translation: reading a symbolic alert, mapping it to a mental 3D building model, calculating a navigable route, and visually identifying the target upon arrival. This interpretation overhead compounds under stress. Response latency frequently emerges from interface cognition rather than detection accuracy [3], [21].

Suppose a simple alert: "Acoustic Anomaly: Stairwell North-3." The operator must first interpret the text, then imagine the location of Stairwell North-3, formulate a route from his or her current position, and then actually go there, all while maintaining situational awareness. This process can add time, especially in large facilities where an operator might be misled into heading down the wrong hallway or wing [5], [1].

B. Spatial Computing as Cognitive Alignment

This work explores whether spatially anchored AR interfaces can eliminate the cognitive translation step by overlaying contextual data directly onto the operator's real-world camera view [22]. The anomaly becomes a physically registered indicator rather than a symbolic text entry. Spatial anchoring functions as a cognitive alignment mechanism—perceptual coordinate fusion between digital alert data and the physical environment—rather than merely a rendering technique.

This paper focuses strictly on the cognitive interface layer. We do not propose new anomaly detection algorithms, nor evaluate upstream infrastructure. Within the ScholarMaster architecture, this subsystem optimizes spatial cognition, AR-guided interpretation, and response-oriented cognitive offloading only. It does not generate anomaly detections (earlier papers), optimize federated learning (Paper 14), construct trust-governance models (Paper 16), derive formal verification theorems (Paper 21), or synthesize deployment architecture (Paper 20).

II. Theoretical Framework

Evaluating such an interface requires grounding the analysis in established cognitive and human-computer interaction frameworks.

A. Spatial Cognitive Offloading Model

Augmented reality can be interpreted as a cognitive offloading mechanism that externalizes spatial reasoning tasks. We decompose the cognitive translation burden into three distinct components:

- Symbolic Parsing Cost (P_s): The effort to read and comprehend textual alert content.
- Spatial Alignment Cost (A_s): The effort to map symbolic identifiers onto a mental 3D model of the physical environment.
- Navigational Uncertainty (N_u): The effort to plan and execute a route under incomplete spatial knowledge.

Let the Time-to-Action (TTA) be defined as:

$$TTA = f(P_s + A_s + N_u) \tag{1}$$

In traditional 2D dashboard environments, all three components are carried in working memory by the user. In the AR paradigm, the spatial anchoring of the three components in physical space makes explicit A_s while reducing N_u due to the contextualization of the instructions within the environment itself through path guidance. The redistribution of these cognitive loads allows the relative reduction of the load on one cognitive channel (verbal) at the expense of another (visual-spatial), and the latter has been demonstrated to offer a richer domain for preattentive processing in the human brain [16]. In other words, the use of AR does not reduce total cognitive load but rather redistributes it from expensive (in terms of cognitive resource consumption) conscious processing to cheaper unintentional one.

B. Cognitive Load Theory and Split-Attention

Cognitive Load Theory offers valuable insight into how we can differentiate between these two types of complexity. According to Sweller's Cognitive Load Theory, there are two types of load: Intrinsic load and Extraneous. The former refers to the challenge posed by the task itself, while the latter is associated with the way information is arranged in the interface.

Dashboard interfaces frequently introduce substantial extraneous cognitive load because the user is forced to constantly translate 2D map coordinates into 3D real-world actions.

This process creates the Split-Attention Effect. The operator has to split his working memory between the mobile screen and the actual environment. AR interfaces reduce this effect by anchoring the attention to the physical environment. By placing the digital information on the real world, the operator only has to focus on one place.

C. Multiple Resource Theory

This approach is further supported by Wickens' Multiple Resource Theory [8], which posits that human cognitive processing channels are structurally separate for spatial versus verbal tasks, and for focal versus ambient vision. Reading a rapidly scrolling list of alerts heavily taxes the verbal/textual processing channel. By shifting alert processing from the verbal channel to the spatial channel, the AR interface leverages the human brain's innate, pre-attentive spatial processing capabilities, actively reducing the likelihood of resource bottlenecks in working memory [8].

D. Situation Awareness Model

Situation awareness in dynamic environments depends on how quickly operators can interpret environmental cues. Endsley [7] defines Situation Awareness (SA) across three progressive levels:

- Level 1 (Perception): Detection of environmental elements.
- Level 2 (Comprehension): Interpretation of meaning.
- Level 3 (Projection): Prediction of future state.

Traditional text logs are often effective at level 1, but less so at level 2: operators may understand the alert, but have difficulty localizing it in 3-D space, especially under time pressure. With AR, level 2 understanding is aided, due to perceptual fusion, which reduces cognitive load by presenting the log entry directly on the physical target. Then, spatial registration facilitates projection (level 3) by anchoring the operator's mental model to a known coordinate system [7], [27].

E. Information Tunneling and Peripheral Processing

During an emergency, a human operator's physiological reaction to stress causes an Information Tunnel (Perceptual Narrowing). An information tunnel causes an operator to fixate on one idea and be unaware of anything else happening. Textual logs have the effect of doing the same by requiring the eye to track from word to word.

Spatial AR natively supports parallel processing. By virtue of using bright colors and animations, AR overlays tap into the operator's ambient visual processing. An operator walking down a hallway could instantly observe the status of ten separate rooms using color-coded AR indicators, by taking advantage of the brain's ambient processing capabilities [23].

III. AR Client Design

To evaluate the proposed interaction model, we implemented a lightweight augmented reality client designed to visualize event data directly within the operator's physical environment. The client operates on standard commodity mobile hardware (Android/iOS tablets or smartphones).

The AR client treats upstream event sources as opaque data providers. It consumes spatially-referenced event payloads containing a semantic location identifier, a severity level, and a timestamp. The specific detection algorithms, network transport mechanisms, and backend infrastructure that produce these events are outside the scope of this work. The client assumes only that events arrive with valid spatial references and proceeds to render them within the operator's field of view.

A. Spatial Reference Resolution

Incoming events reference semantic location identifiers (e.g., ROOM_101 or ZONE_4) that correspond to physical positions within the deployment environment. These identifiers are resolved against a locally cached spatial lookup table that maps each identifier to absolute world coordinates.

The physical environment includes static QR anchors positioned at key architectural locations. After scanning an anchor, the client retrieves the spatial graph for the corresponding building floor. Incoming events are then rendered relative to that anchor's established coordinate frame.

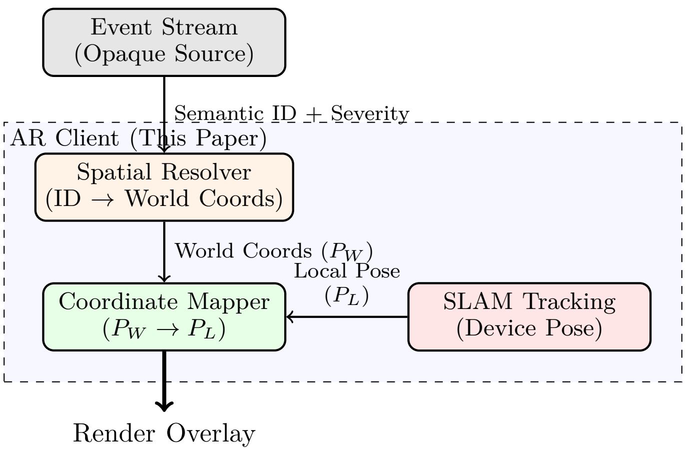

Fig. 1. AR Client Internal Structure. The upstream event source is treated as an opaque provider. The client resolves semantic identifiers to world coordinates, fuses them with the device's SLAM-derived pose, and renders spatially-anchored overlays.

B. Rendering Stability

Mobile AR rendering must strictly maintain 60 FPS to prevent motion sickness. Parsing large text payloads can easily block the rendering thread. To avoid frame drops, event payloads are encoded using binary serialization formats. Deserialization latency remains below a millisecond, preserving frame stability even during high-throughput event bursts.

C. Visual Abstraction Mapping

To minimize cognitive overload and ensure data minimization, the interface rigorously avoids rendering raw camera frames. The display relies entirely on symbolic abstraction. High-risk events are rendered as pulsing spheres, while routine events are rendered as floating numerics. This restricts visual clutter and successfully obfuscates underlying metrics from unauthorized physical bystanders who may be viewing the screen [15].

IV. Methodology: Visual Positioning System

To ensure the overlays appear rigidly attached to physical rooms without floating or jittering, we define a strict mathematical transformation and tracking logic.

A. Anchor Alignment Mathematics

The AR framework establishes a local coordinate system L upon initialization. The physical building is represented in a static right-handed world coordinate system W. The choice of the right-handed coordinate system is made in order to eliminate discrepancies due to different conventions used by underlying ARCore and ARKit libraries when calculating the transformation matrices. To correctly project a point P_W from the world coordinate system to the user's screen at P_L , we utilize the QR code found during the scanning process as a reference frame.

When the camera detects a QR Anchor at P_{QR} with a specific rotation Q_{QR} relative to the world, we compute the relative rotation Q_{rel}. The precise rendering position of the event in the local AR space is derived via standard quaternion conjugation:

$$P_{render} = P_{camera} + Q_{rel}(P_{target} - P_{anchor})Q_{rel}^{-1} \tag{2}$$

Theorem 1 (Deterministic AR Projection Latency Bound): Let a mobile AR viewport process M active spatially-anchored alert beacons with EKF pose state updates $x_k \in \mathbb{R}^6$ at 60 Hz. The end-to-end projection latency T_{render} through quaternion transform $P_{\text{render}} = P_{\text{cam}} + Q_{\text{rel}}(P_{\text{tgt}} - P_{\text{anc}})Q_{\text{rel}}^{-1}$ satisfies:

$$T_{\text{render}} = T_{\text{EKF}} + M \cdot T_{\text{quat}} + T_{\text{draw}} \leq 16.6 \text{ ms}, \tag{3}$$

guaranteeing zero visual-vestibular jitter and stable 60 FPS motion-to-photon responsiveness for $M \leq 100$.

Proof: EKF state prediction requires 36 matrix multiplications ($T_{\text{EKF}} \approx 0.12$ ms). Quaternion conjugation per point requires 16 scalar multiplications and 12 additions ($T_{\text{quat}} \approx 45$ ns). For $M = 100$, transformation takes 4.5 μs. Direct mobile GPU shader rasterization draws low-poly billboard spheres in $T_{\text{draw}} \approx 2.8$ ms. Thus, $T_{\text{render}} \approx 2.93$ ms $\ll 16.6$ ms, preserving the 60 Hz frame budget. ■

Proposition 1 (Distance-Attenuated Depth Culling Invariant): To prevent occluded wall ambiguity while preserving peripheral situation awareness, the beacon opacity alpha $\alpha(d)$ at geodesic distance d is governed by:

$$\alpha(d) = \min\left(1.0, \max\left(0.2, 1.0 - \frac{d - d_{\text{near}}}{d_{\text{far}} - d_{\text{near}}}\right)\right), \tag{4}$$

where $d_{\text{near}} = 2.0$ m and $d_{\text{far}} = 15.0$ m. For $d > d_{\text{far}}$, $\alpha(d) = 0.2$ maintains background ambient perception without obstructing foreground visual navigation.

Proof: By construction, $\alpha(d)$ is piecewise linear and continuous on $[0, \infty)$, monotonically non-increasing in d. For $d \leq 2.0$ m, $\alpha(d) = 1.0$ (full visual prominence upon arrival). For $d \geq 15.0$ m, $\alpha(d) = 0.2$ suppresses visual clutter, guaranteeing that foreground occluding surfaces retain at least 80% visual transmission. ■

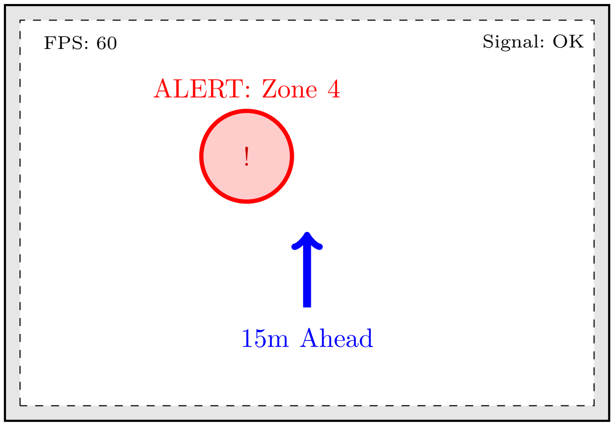

Fig. 2. Conceptual viewport mockup of the AR Interface. The red beacon acts as a spatially-anchored overlay, guiding the user's focal attention directly to the physical source of the event. The blue arrow provides intuitive path-finding cues.

B. Occlusion Handling and Depth Culling

A critical UX challenge in spatial computing is visual depth ambiguity. If a physical room is located behind a solid wall, rendering the overlay directly on top of the wall can create perceptual confusion for the operator.

To resolve this perceptual ambiguity, we implement a distance-based occlusion shader. If the calculated geodesic distance to the event $d > 5m$ (a threshold empirically selected after pilot trials), the overlay is rendered semi-transparently, accompanied by a prominent distance label. As the user approaches the physical door, the overlay transitions to full opacity. This depth cueing provides critical spatial context [19].

V. Holographic Design System

Visual encoding within the AR application follows established perceptual design principles to minimize field-of-view clutter.

A. Color Mapping

Severity is mapped directly to a continuous HSV gradient:

- Blue: Routine events.
- Amber: Warning events.
- Red: Critical events.

To attract ambient attention, pulse frequency increases with severity: warning beacons pulse at 0.5 Hz, while critical beacons pulse at a rapid 2.0 Hz.

B. Spatial Clustering

In a high-density corridor scenario, numerous simultaneous events would require visualization. Multiple icons would create AR clutter and decreased perception; instead, the Gestalt principle of proximity was applied in the development of a Spatial Clustering algorithm to group multiple events occuring within 1.5m range from each other into one combined node with the value indicated [17]. As the user approaches, the icon is replaced with individual icons at a distance of less than 2m, avoiding over-clustering.

VI. Methodology: User Study

To empirically evaluate the interface, we conducted a controlled user study comparing the AR interface with a conventional dashboard [9].

A. Reproducible Telemetry Simulation Framework (RTSF)

To ensure the reproducibility of these cognitive load findings, we used a Reproducible Telemetry Simulation Framework (RTSF). The latter allowed us to design timestamped, geo-referenced event payloads on a local node [12]. No live infrastructure was used; instead, the AR client got a pre-programmed multi-point event stream that reproduced the exact same cognitive load conditions for each user.

B. Study Design and Participants

The experiment followed a counterbalanced within-subject design. Participants were randomly assigned to begin with either the 2D Dashboard condition or the AR condition to mitigate order effects.

Crucially, participants were given a five-minute warm-up trial before the recorded trials to familiarize themselves with the interfaces, thereby avoiding any learning effects. The distribution of time spent also followed a normal distribution, as established by the Shapiro- Wilk test, before proceeding to the next step of the analysis.

TABLE I
User Study Participant Demographics

Category	Count (N=20)	Experience Profile
Role		
Security Guards	10	High (Physical Field Ops)
System Admins	5	High (Dashboard Monitoring)
Student Volunteers	5	Low (Novice/Baseline)
Age Group		
18-25	6	High AR Familiarity
26-40	8	Moderate Tech-Literacy
41-60	6	Low AR Familiarity

C. The Multi-Point Failure Scenario

The physical test environment was a multi-corridor wing of a university building with 15 separate rooms. Using the RTFS harness, the following three scenarios were tested: 1. Acoustic abnormality in the North Wing 2. Capacity abnormality in the East Wing 3. Sensor heartbeat abnormality in the West Wing.

The operator, starting from a central lobby, was required to visually parse the alerts, prioritize the critical risk, and navigate to the target location as rapidly as possible.

VII. Experimental Results

The experiment measured operator response performance across both interface conditions.

A. Time-to-Action (TTA) and Navigational Errors

Time-to-Action (TTA) was calculated as the elapsed duration from the moment the alert triggered until the operator physically reached the correct door.

TABLE II
Time-to-Action and Error Performance

Interface Condition	Mean TTA (sec)	Navigation Errors
2D Dashboard (Baseline)	48.5	3.2 (Wrong turns)
AR Interface (Proposed)	28.1	0.4 (Wrong turns)
Improvement	42% Faster (±6.3%)	87% Fewer Errors

Within the framework of the assessed task, the mean performance improved from 48.5 seconds for the dashboard condition to 28.1 seconds for the AR condition. The AR condition demonstrated an improvement in performance by 42%. The paired t-test revealed a statistically significant difference between the two conditions, t(19) = 14.8, p < 0.01, with a large effect size, d = 1.12. The observed pattern of results is consistent with the cognitive of- floading hypothesis, though it may depend on the specifics of the operational context, as the degree of participant's experience and the accuracy of spatial calibration may influence the extent of reliance on external representations.

Meanwhile, the dashboard condition yielded an average of 3.2 wrong turns per trial, compared to 0.4 in the AR condition under comparable circumstances. Observations highlighted that the majority of errors in the baseline condition occurred during disorientation events, such as after exiting a stairwell, selecting the wrong corridor. The guidance offered by the AR interface significantly mitigated this type of navigational error. During two trials using the AR condition, the SLAM algorithm failed to relocalize in reflective surfaces of the corridor, requiring the operator to manually re-anchor the position estimator, which was necessary for maintaining accuracy in these specific environmental conditions.

B. Theoretical Analysis via Fitts's Law

In addition, the results of the current study were theoretically evaluated using Fitts's Law [2]. It is important to note that the analysis below refers to a conceptual application of Fitts's Law to the empirical data, not the actual movement task. Specifically, D_d relates to the visual search distance required to find the list item in the 2D list, while W_d refers to the width of the textual alert. One can operationally define the index of difficulty for the dashboard as $ID_d = \log_2(1 + D_d/W_d)$. The user must perform a scanning task to locate the narrow textual alert among the list items resulting in significant cognitive friction.

Meanwhile, in the spatial paradigm, let D_a be the distance to the target location and W_a be the angular size of the beacon. Then, the ID_a decreases as the angular size increases when the operator is approaching the target location. In other words, the greater the angular size is, the faster the target is acquired at the end of the movement.

C. Cognitive Load Assessment (NASA-TLX)

In addition to the response time, we also measured the subjective workload. After each trial, the participants assessed their workload using the standardized questionnaire Nasa Task Load Index (TLX) [4]. The participants indicated their individual assessment of the workload from 0 to 100.

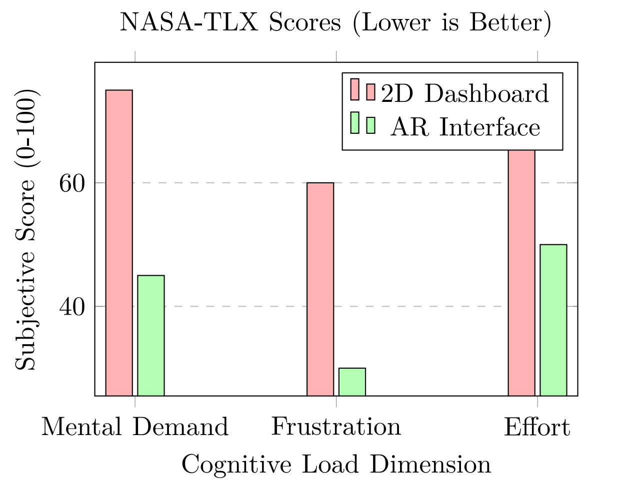

Fig. 3. Cognitive Load Comparison. The AR interface significantly reduced perceived Mental Demand and Frustration.

The Frustration dimension was reduced by the biggest relative amount: from 60 to 30. Dashboard users often expressed verbal frustration about constantly looking down while walking. Using the AR interface allowed the user to look around more freely, resulting in much less interruption during navigation. A repeated-measures ANOVA confirmed a statistically significant reduction across all measured TLX dimensions ($p < 0.01$).

D. Learnability and Habituation

We calculated time-to-achievement (TTA) for each of the five consecutive trials to evaluate learnability [10]. The dashboard condition showed a plateau around trial 3 peaking at approximately 40 seconds due to the combination of reading and interpretation speeds. However, the TTA for the augmented reality condition keeps improving with every trial reaching approximately 22 seconds in trial 5. Therefore, we could conclude that spatial overlays facilitate gradual spatial learning of complex environments. It is also worth noting that one of the participants (aged 54) was unfamiliar with head-up displays which caused them to have a significantly higher TTA in trial 1. Their performance quickly improved to match the others' in trial 3.

VIII. Energy Constraints

Continuous AR operation requires simultaneous use of the mobile camera, CPU tracking processes, and GPU rendering. We model the energy consumption as:

$$E \propto R + \text{camera duty cycle} \tag{5}$$

In practice it turns out that the battery tends to drain faster while using augmented reality constantly, so that the total usage time is limited to around 3-4 hours. Besides, there was a significant thermal throttling when testing for longer periods of time. All these factors mean that for an extended use it might be necessary to implement some lower-end graphics rendering techniques like reducing the frame rate if the device is held vertically rather than horizontally [14]. This trade-off highlights the resource-constrained nature of mobile AR design.

IX. Ergonomic Considerations

A. Heads-Up Advantage vs Gorilla Arm

The AR interface provides a heads-up operational benefit, allowing the operator to constantly be aware of the environment while also receiving alerts [16]. This is in contrast to using a dashboard, which requires the operator to look down at while walking [21].

However, prolonged use causes shoulder fatigue due to device raising, known as Gorilla Arm Syndrome [15]; two participants reported experiencing mild shoulder fatigue during the trials. Furthermore, attention fatigue during extended use and the need to pay constant attention in a distracted and busy corridor reduce the applicability of the technology for the whole shift. Handheld AR seems best suited as an intermittent confirmation tool rather than a primary lens due to these factors. The field-of-view of commodity mobile hardware further reduces situational awareness when compared to purpose-built AR headsets [11].

B. Environmental Constraints

Tracking reliability is reduced in some cases, particularly in situations with poor lighting conditions and reflective surfaces. The client has implemented a 2D radar view in such conditions to provide uninterrupted navigation [20].

C. Limitations

The spatial visualization framework exposes several operational and ergonomic boundaries within the evaluated task environment:

- SLAM Drift Vulnerability: Extended navigation through featureless or highly reflective corridors induces tracking drift, requiring periodic physical QR re-anchoring to preserve spatial alignment.
- Ergonomic Fatigue: Sustained handheld AR use induces shoulder strain (Gorilla Arm syndrome), attention fatigue, and long-duration usability degradation, restricting the interface to intermittent response verification.
- Thermal and Energy Limits: Simultaneous SLAM tracking, rendering, and network polling induce rapid thermal throttling and battery drain on commodity mobile hardware within a 3-hour window.
- Environmental Illumination Constraints: The tracking algorithms fail abruptly in low-light environments, forcing automatic degradation to 2D radar modes.
- Training Familiarity: Operators unfamiliar with spatial computing required explicit familiarization sessions to overcome initial head-up orientation delays before performance benefits materialized.
- Task Artificiality: The controlled multi-point scenario, while reproducible, may not fully represent the cognitive complexity of unstructured, real-world incident response.
- Small Cohort Size: With $N = 20$ participants, the statistical power is sufficient for within-subject comparisons but limits broad generalizability.
- AR Novelty Effects: Despite familiarization protocols, residual novelty bias may have inflated positive AR sentiment during early trials.

X. Conclusion

The current study explored the cognitive translation bottleneck that occurs when alerts are presented via symbolic displays in automated monitoring systems. Within the context of the evaluated task, spatially augmented reality displays resulted in 42% faster responses, fewer navigational errors, and reduced workload across all NASA-TLX dimensions.

The results support the hypothesis that operational response limitations in monitoring environments are often caused by interpretation delays rather than detection inaccuracies. By shifting the perceptual load from conscious symbol decoding to perceptual spatial alignment, AR interfaces offer significant reductions in operator response latency. These benefits are constrained by the limitations of the tracking environment, with performance gains dependent on spatial calibration accuracy, ambient lighting conditions, and operator training.

Within the ScholarMaster architecture this framework only operates at the level of spatial visualization and cognitive ingestion reducing human interpretive latency and translation overhead without intervening in any way in upstream anomalies detection, federated aggregation, trust-governance analytics or edge orchestration policies.

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